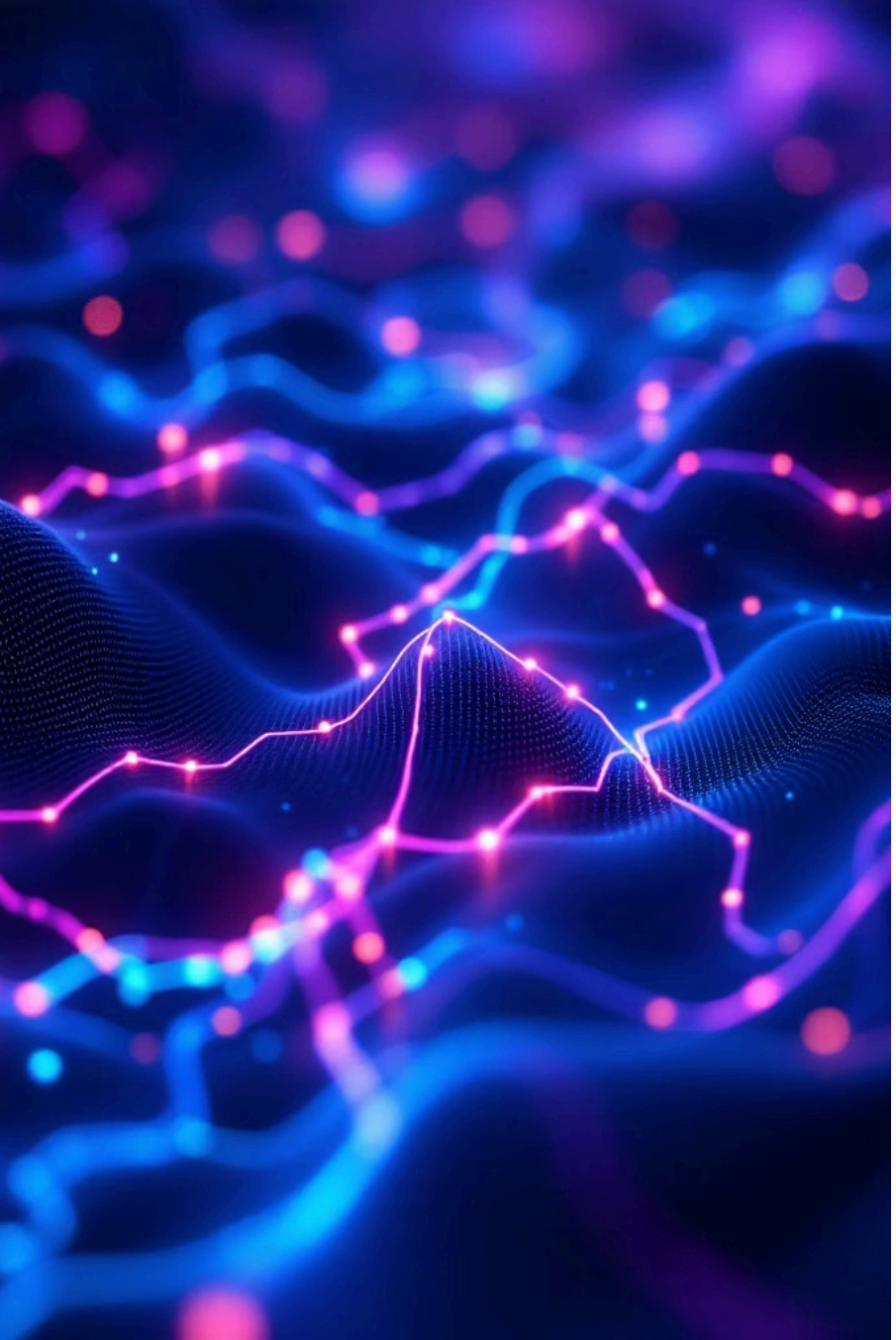

Customer Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.

Made with GAMMA

Project Overview

Goal

Analyze 3,900 purchases to optimize operations.

Focus

Spending patterns, segments, product preferences, subscriptions.

Impact

Inform strategic business decisions.

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Dataset Summary

Data Points

- 3,900 Rows
- 18 Columns

Key Features

- Customer demographics
- Purchase details
- Shopping behavior

Missing Data

37 values in 'Review Rating'.

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Exploratory Data Analysis (Python)

01

Data Loading & Exploration

Imported with `pandas`, checked structure and stats.

02

Missing Data Handling

Imputed 'Review Rating' with median by category.

03

Column Standardization

Renamed to `snake_case` for readability.

04

Feature Engineering

Created `age_group` and `purchase_frequency_days`.

05

Database Integration

Loaded cleaned data to PostgreSQL for SQL analysis.

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SQL Analysis: Key Business Insights

Revenue by Gender

Male: $157,890

Female: $75,191

Top 5 Products by Rating

- Gloves (3.86)
- Sandals (3.84)
- Boots (3.82)

High-Spending Discount Users

839 customers spent above average with discounts.

Shipping Type Comparison

Express: $60.48 avg.

Standard: $58.46 avg.

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SQL Analysis: Deeper Dive

Subscribers vs. Non-Subscribers

Subscribers: 1,053 customers, $62,645 revenue.

Non-subscribers: 2,847 customers, $170,436 revenue.

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%).

Customer Segmentation

Loyal: 3,116

Returning: 701

New: 83

Repeat Buyers & Subscriptions

958 repeat buyers are subscribers.

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Top Products & Age Group Revenue

Top 3 Products per Category

- Accessories: Jewelry, Sunglasses, Belt
- Clothing: Blouse, Pants, Shirt
- Footwear: Sandals, Shoes, Sneakers
- Outerwear: Jacket, Coat

Revenue by Age Group

- Young Adult: $62,143
- Middle-aged: $59,197
- Adult: $55,978
- Senior: $55,763

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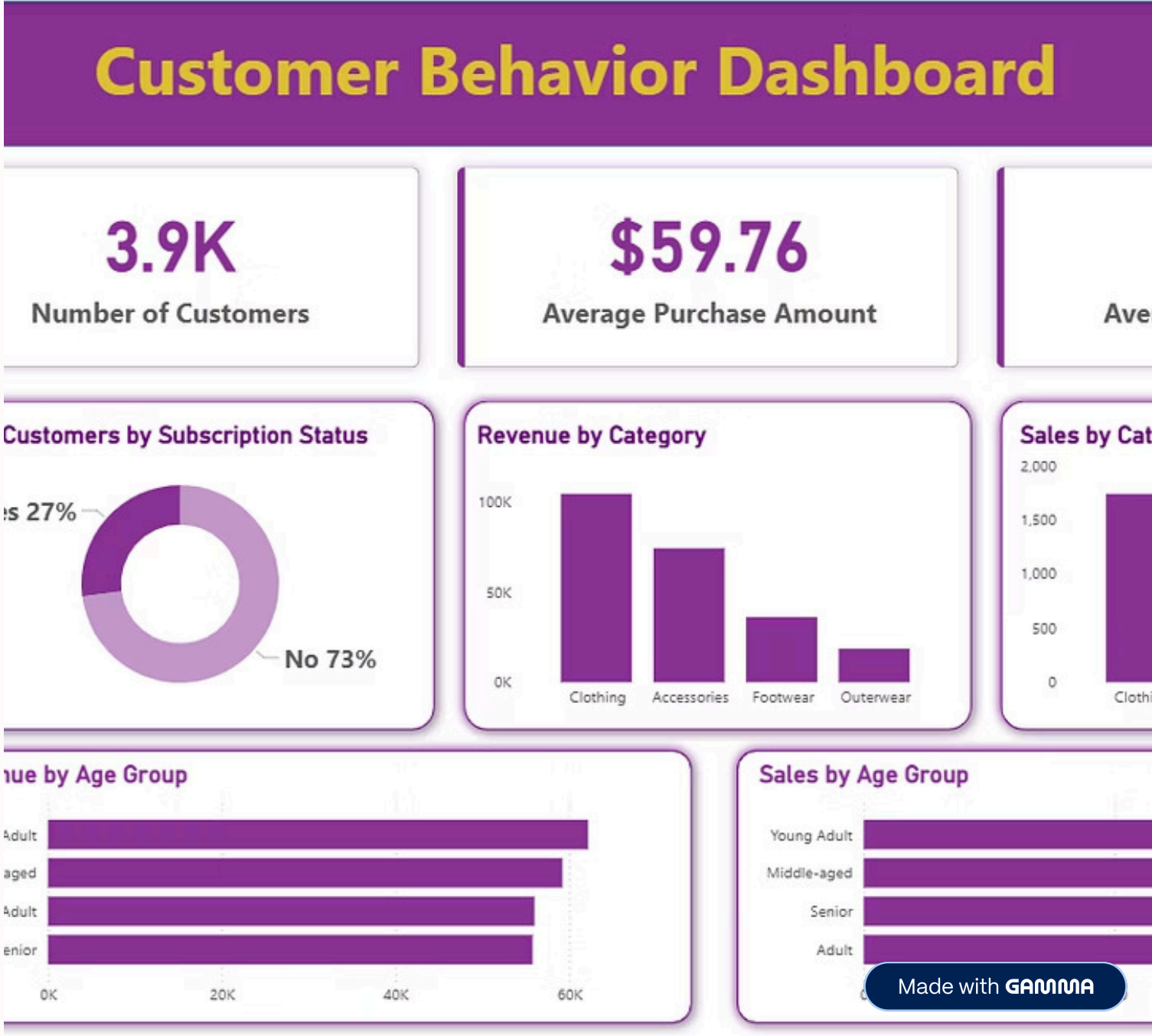
Power BI Dashboard

Interactive dashboard for visual insights.



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Key Metrics & Trends

Overall

- Customers: 3.9K
- Avg. Purchase: $59.76
- Avg. Review: 3.75

Subscription Status

27% Yes | 73% No

Revenue by Category

- Clothing: $120K
- Accessories: $100K
- Footwear: $50K
- Outerwear: $20K

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits.

Customer Loyalty

Reward repeat buyers.

Review Discount Policy

Balance sales with margins.

Product Positioning

Highlight top-rated items.

Targeted Marketing

Focus on high-revenue age groups.

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